DHARNISH GIRI

Robotics & AI Engineer · Physical AI · Teleoperation · ROS2 · Deep Reinforcement Learning · MPC
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EDUCATION

M.Sc. Automation and Robotics Oct 2024 – Present
Technical University of Dortmund, Germany

- Relevant Coursework: Deep Reinforcement Learning · Machine Learning Methods for Engineers

B.Eng. Robotics and Automation Aug 2019 – Apr 2023
PSG College of Technology, India

TECHNICAL PROJECTS

STORM – Salient TeleOperated over Resilient Mobile Networks Nov 2025 – Jun 2026
ROS2 Jazzy · CARLA Simulator · Python · C++ · Zenoh DDS · GStreamer · Docker | Chair for Communication Networks, TU Dortmund

- Architected a modular ROS2 Jazzy sim-to-sim control bridge coupling CARLA's full vehicle dynamics simulation with a Fanatec force-feedback hardware-in-the-loop interface, enabling closed-loop human-in-the-loop vehicle control over simulated mobile network channels.
- Built a multi-camera H.265 perception pipeline (GStreamer + nvh265enc) with adaptive bitrate and resolution degradation tied to real-time network conditions.
- Implemented Zenoh middleware (rmw_zenoh_cpp) for resilient, low-latency command and telemetry streaming across a two-PC networked architecture, with all components fully containerised via Docker CE and NVIDIA Container Toolkit.

Franka Pick-and-Place via Hybrid RL + VLA May 2026 – Jun 2026
Python · MuJoCo · Stable-Baselines3 · SAC · SmolVLA · Gymnasium

- Designed a custom Gymnasium environment (FrankaPickPlaceEnv) wrapping the MuJoCo Menagerie Franka Emika Panda model, with a Dict observation space encoding 7-DOF joint positions, joint velocities, gripper pose, box pose, and relative gripper-to-box vector.
- Implemented a hybrid control architecture: a Soft Actor-Critic (SAC) policy handles the approach phase — navigating the arm from free space to near-proximity of the pick target — using a shaped reward combining box-to-target distance minimisation and gripper-to-box proximity for dense learning signal.
- Integrated SmolVLA (HuggingFace's compact Vision-Language-Action model) for grasp execution: once SAC brings the end-effector into proximity, SmolVLA handles the contact-rich grasping phase, decoupling the high-level navigation problem (well-suited to RL) from the fine manipulation problem (well-suited to VLA).
- Trained SAC with MultiInputPolicy over 300k environment steps (buffer size 1M, batch size 256), leveraging off-policy sample efficiency for continuous 8-DOF action control; structured observations with achieved_goal and desired_goal fields for HER compatibility.

Gesture-Controlled Dual-Arm Manipulation System Jan 2026 – Feb 2026
ROS2 Jazzy · MoveIt2 · MediaPipe · OMPL · Kalman Filtering

- Built a real-time bilateral teleoperation pipeline using MediaPipe Hands for vision-based operator control of a dual UR5 + Robotiq 85 system, eliminating physical controllers — directly relevant to physical AI human-robot interaction and imitation learning data collection.
- Implemented Kalman filtering on 2D hand landmark streams to suppress sensor noise before Cartesian pose computation, enabling smooth, continuous joint-space trajectory execution via MoveIt2's GetPositionIK service.
- Authored full URDF/SRDF system model including collision matrices, planning group definitions, and end-effector configurations for zero-configuration MoveIt2 deployment of the dual-arm platform.

Autonomous Highway Navigation via Model Predictive Control Jun 2025 – Jul 2025
Python · CasADi · Nonlinear Vehicle Dynamics · Real-Time Optimisation

- Derived a nonlinear kinematic bicycle model and implemented a receding-horizon MPC controller from scratch in CasADi for multi-lane highway navigation, achieving real-time solve rates exceeding 20 Hz on CPU.
- Designed cost functions covering trajectory tracking, obstacle proximity penalties, and input rate minimisation with hard constraints on velocity (≤120 km/h) and lane boundaries.
- Validated robustness across 5+ dynamic cut-in scenarios at high speed with zero collision incidences.

PROFESSIONAL EXPERIENCE

Research Assistant | Lehrstuhl für Produktionssysteme (LPS), TU Dortmund Mar 2026 – Present

- Designing and maintaining a PostgreSQL research database for ExoExpert, a mechanical exoskeleton development project, managing structured storage and retrieval of biomechanical measurement data.
- Building queryable data pipelines between experimental measurement sessions and analysis tooling, enabling data-driven iteration on exoskeleton control parameters.

Robotics Engineer | Titan Engineering and Automation Limited, India Aug 2023 – Aug 2024

- Programmed and commissioned ABB and Mitsubishi 6-DOF industrial manipulators for production automation cells, developing collision-free motion programs and I/O-based PLC integration for end-to-end line control.
- Achieved 15% cycle time reduction on a robotic weld cell through trajectory re-optimisation and dynamic parameter tuning, validated against real-world kinematic constraints and live sensor feedback.
- Conducted field commissioning and acceptance testing, systematically validating robot motion programs against physical measurement data.

Robotics Intern | Titan Engineering and Automation Limited, India Feb 2023 – Jul 2023

- Developed motion programs for 6-DOF industrial arms across pick-and-place and palletising applications; hands-on experience with robot kinematics, TCP calibration, and safety-zone configuration.

TECHNICAL SKILLS

Programming: Python (NumPy, SciPy, CasADi, OpenCV, MediaPipe, Stable-Baselines3), C++ (ROS2 nodes), MATLAB

AI & Learning: Deep Reinforcement Learning (SAC, PPO), Vision-Language-Action Models (SmolVLA), Imitation Learning, MuJoCo, Gymnasium, Stable-Baselines3

Simulation: ROS2 (Jazzy / Humble), CARLA, MuJoCo, Gazebo, Nav2, MoveIt2, URDF/SRDF

Control: Model Predictive Control (MPC), nonlinear vehicle dynamics, trajectory optimisation, Kalman filtering, IK resolution

Middleware: DDS, Zenoh (rmw_zenoh_cpp), ROS2 topic/service/action interfaces, GStreamer, real-time systems

DevOps: Git, Docker, NVIDIA Container Toolkit, Ubuntu/Linux, SSH, VS Code

Industrial: ABB, KUKA, Mitsubishi, Universal Robots

LANGUAGES & AVAILABILITY

English: Fluent (IELTS 7.0) | German: B1 (in progress) | Tamil: Native

Availability: Available from August 2026. Currently enrolled M.Sc. student at TU Dortmund — flexible on weekly hours.